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DEVELOPMENT OF A POTENTIAL DISTRIBUTION MODEL FOR THE HYACINTH MACAW (*ANODORHYNCHUS HYACINTHINUS*)

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Summary

This study aimed to model the potential distribution of the Hyacinth Macaw (Anodorhynchus hyacinthinus) across its vast range in central and eastern South America, addressing the urgent problem of habitat loss and insufficient protected areas that threaten the species' survival. By employing advanced species distribution models that integrate multi-temporal occurrence records and land use/cover data, it is possible to assess changes in suitable habitat over the past 25 years and identified key environmental predictors influencing macaw distribution. The results revealed that while the overall modelled range of the Hyacinth Macaw expanded by approximately 75,000 km^2, only 8–10% of this habitat currently falls within protected areas, and more than 20,500 km^2 of protected land has been degazetted in recent decades. The most influential factors explaining habitat suitability were the distribution of key palm and tree species, forest cover, and recent changes in land use, particularly the expansion of agriculture and pasture. These findings highlight the dynamic and regionally variable response of the species to land cover change, with expansion in the Amazon, stability in the Pantanal, and fragmentation in the Cerrado. The distinctive feature of this research is the use of multi-temporal modelling, which allowed to capture spatiotemporal patterns and identify priority areas for conservation and restoration, even in data-poor and remote regions. The models developed here can be practically applied to guide conservation planning, inform protected area designation, and support community-based monitoring initiatives across the Hyacinth Macaw's range.

Keywords: Hyacinth Macaw, species distribution model, habitat loss, land use change, protected areas, Pantanal, Cerrado, Amazon, conservation planning, remote sensing.

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1. Introduction

In recent years, many psittaciform species have faced various threats to their existence, with more than 40 % classified as endangered or near-threatened [1]. Among the numerous factors affecting these species, notable threats include wildlife trafficking [2], habitat loss due to logging and agricultural activities [3], and human intrusions [4]. The hyacinth macaw (*Anodorhynchus hyacinthinus*) is the largest bird in the Psittacidae family and, consequently, the largest macaw in the world. It can weigh up to 1.3 kg and measure over a meter in length from beak to tail tip [5]. In Brazil, it occurs in the North, Northeast, and Central-West regions, with apparently isolated populations. This species is non-migratory, monogamous, and highly specialized in its feeding and nesting sites. Some aspects of its reproductive biology and life history have only recently been discovered [6], making its distribution even more sensitive to the aforementioned disturbances.

Understanding species distribution is a global priority for mitigating the environmental pressures of human activities [7]. In this context, species distribution models (SDMs) are valuable tools. SDMs are numerical methods that combine species occurrence or abundance data with environmental estimates. They are widely used in ecology and conservation to predict species distributions [8]. The core idea is to obtain a continuous spatial gradient of habitat suitability, using environmental variables as predictors and species presence (0 – absent, 1 – present) as a binary response variable. This approach allows for the identification of areas that are more or less suitable for the occurrence of a given species.

The aim of this study is to develop a potential distribution model for *Anodorhynchus hyacinthinus* in Brazil, based on environmental variables associated with occurrence records. This research seeks to contribute to the theoretical understanding of the species' ecology by identifying key environmental factors influencing its presence, as well as providing insights for management and conservation efforts.

2. Methods

Occurrence points were identified for *Anodorhynchus hyacinthinus* by collecting data available from GBIF (www.gbif.org), obtaining 13,847 occurrence records for Brazil. Then the duplicate records and those with missing geographic coordinates were removed, resulting in 2,591 records for Brazil.

For environmental variables, let's use two categories: bioclimatic and topographic, all extracted from the Bioclim database (www.worldclim.org) at a 30-second resolution. Let's conduct collinearity analyses using Pearson's correlation coefficient and the Variance Inflation Factor (VIF) metric to identify potential collinearity issues among the variables. Of the 20 environmental variables collected, 13 were eliminated due to multicollinearity issues (VIF greater than 10 and high pairwise correlation coefficients).

There are various modeling methods, each with specific algorithmic characteristics aimed at identifying the most accurate potential distribution for species. In this study, let's use Maxent, Bioclim, GLM (Generalized Linear Models), and GAM (Generalized Additive Models). Maxent is a Java-based application that models present and future habitat suitability for species by finding the distribution with maximum entropy [9]. Bioclim calculates a bioclimatic envelope using a digital elevation model (DEM) [10]. The environmental variables of the study area are compared with those of the bioclimatic envelope to predict suitable habitats [11]. GLMs (Generalized Linear Models) are based on parametric functions and can be adjusted for different probability distributions. GAMs (Generalized Additive Models), in contrast, do not use predefined parametric forms. Instead, they apply a selection of smoothing functions along predictor variable gradients to find the best solution [12].

The variability in predictions or projections from different models challenges the reliability of using a single method. A solution is to use multiple models together (ensemble modeling) to enhance prediction robustness [13], which was applied in this study. The dataset was split into two sets: 70% of the data was used to calibrate the models, while 30% was used for testing.

For modeling, let's use R version 4.4.1, utilizing the dismo, rgbif, tidyverse, raster, and sdm packages. Map generation was performed using QGIS 3.34.11.

3. Results

Of the 20 environmental variables collected, 13 had VIF values above 10 and/or Pearson correlation coefficients greater than 0.8 with other study variables. These variables were eliminated to avoid severe multicollinearity issues in model construction. As a result, 7 environmental variables were retained, as described in **Table 1** below.

Table 1

Variables used in the model. A selection criterion of VIF≤10 and Pearson correlation coefficient <0.8 were applied

Variable	VIF
Bio2 (mean diurnal range)	2.22
Bio3 (isothermality)	4.91
Bio5 (maximum temperature of warmest month)	4.98
Bio13 (precipitation of wettest month)	3.84
Bio14 (precipitation of driest month)	8.30
Bio15 (precipitation seasonality – coefficient of variation)	5.18
Elevation	4.44

Among the 7 selected predictor variables, one represents elevation, three represent precipitation (Bio13, Bio14, and Bio15), and the remaining three represent temperature (Bio2, Bio3, and Bio5). Bio2 reflects the average difference between monthly maximum and minimum temperatures throughout the year, while Bio3 measures the degree of temperature fluctuation between day and night.

Among the four models generated, MAXENT exhibited the best performance (AUC = 0.990), followed by BIOCLIM (AUC = 0.986) and GAM (AUC = 0.975) (**Table 2**). The AUC (Area Under

the Curve) is a robust performance metric that represents the area under the curve between the true positive rate (TPR) and the false positive rate (FPR), ranging from 0 to 1. An AUC around 0.5 indicates poor model performance, comparable to random chance, while a high AUC (close to 1) indicates excellent model separability, meaning the model can effectively distinguish and correctly predict locations where the species is present. Other performance metrics used for model comparison are presented in **Table 2**.

Table 2

Performance statistics for the generated models. Based on all evaluation metrics, MAXENT was the best-performing model

Model	AUC	COR	TSS	MCC	F1
GAM	0.975	0.956	0.945	0.953	0.987
GLM	0.969	0.931	0.913	0.932	0.981
BIOCLIM	0.986	0.564	0.933	0.939	0.983
MAXENT	0.990	0.936	0.928	0.914	0.975

The habitat suitability predictions derived from the models are shown in **Fig. 2**. MAXENT predicts an almost entirely continuous and slightly uniform pattern of suitability across Brazil, with no single state standing out significantly. However, the extreme south of Brazil, the far north of the Amazon, and some coastal regions in the Northeast were identified as the least suitable areas for the species (**Fig. 1, *a***).

In contrast, GLM and GAM predict mosaic-like patterns, with higher suitability in the Mato Grosso and Mato Grosso do Sul regions and lower suitability in the South, the extreme north of the Amazon, and along most of the Brazilian coastline (**Fig. 1, *b, c***). BIOCLIM, on the other hand, indicated isolated patches of suitability surrounded by areas of very low suitability (**Fig. 1, *d***).

To generate a more robust suitability prediction, an ensemble model combining all four approaches was applied (**Fig. 2**). The overall suitability pattern appears smoother but less spatially continuous than MAXENT's prediction, with regions of unsuitability consistently surrounding suitable areas. Once again, Mato Grosso and Mato Grosso do Sul emerge as the most suitable regions, while the extreme south of Brazil (Rio Grande do Sul and Santa Catarina) are the least suitable areas.

The southern region of the state of Mato Grosso and the northern region of Mato Grosso do Sul appear as the most suitable areas for the species. The southernmost regions of Brazil, specifically the states of Rio Grande do Sul and Santa Catarina, are identified as the least suitable areas.

Among the variables used in the modeling, the variable Bio3 (isothermality), which measures the degree of temperature fluctuation between day and night, was the most important for the model (**Table 3**), followed by Bio13 (precipitation of the wettest month)**.** The variable Bio14 (precipitation of the driest month) had the least importance.

Table 3

Summary of relative variable importance with percentages representing the importance of the variables based on two metrics: **correlation** and **AUC**

Variable	Correlation	AUC
bio13	24.3%	9.8%
bio14	9.2%	2.1%
bio15	16.5%	9.2%
bio18	15.1%	6.7%
bio2	13.5%	4.6%
bio3	33.6%	23%
bio5	11.5%	5.9%
Elevation	15.9%	7.7%

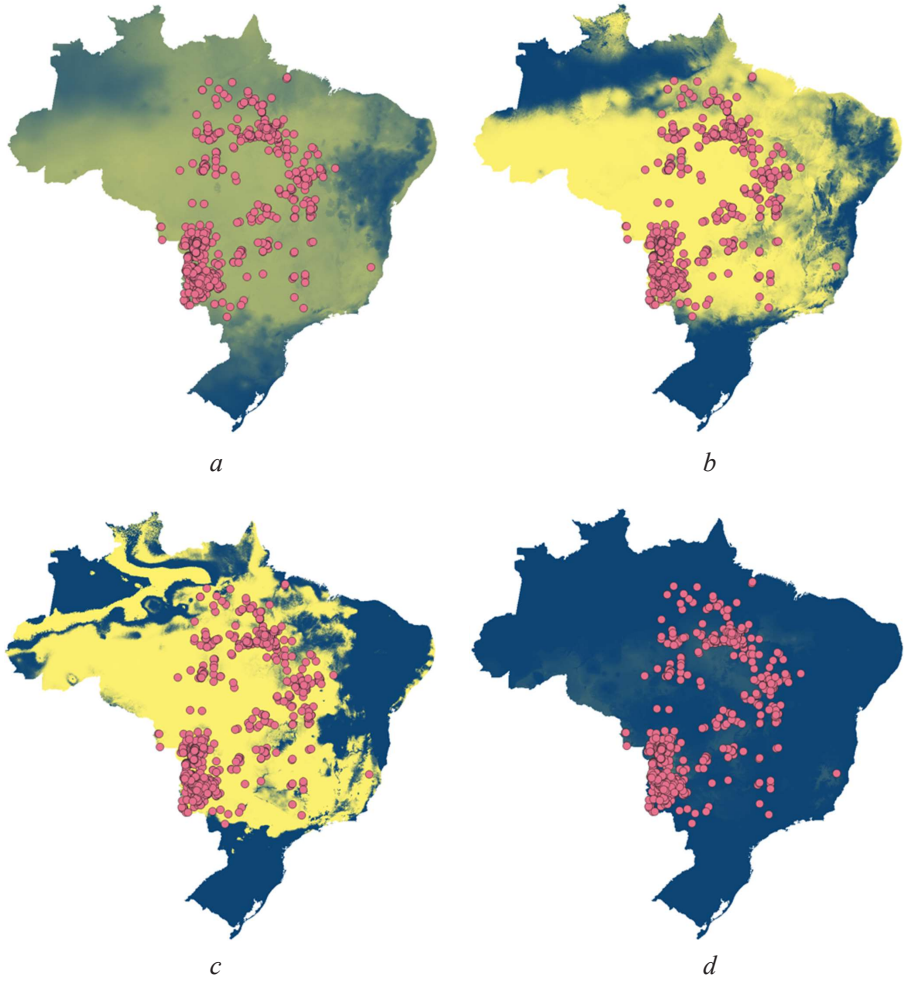


Fig. 1. Habitat suitability for *Anodorhynchus hyacinthinus* according to four species distribution modeling methods. The points represent species occurrence records: *a* – MAXENT; *b* – GLM; *c* – GAM; *d* – BIOCLIM

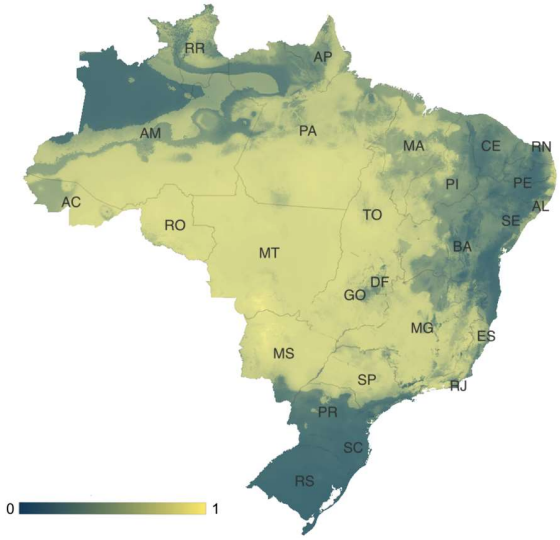


Fig. 2. Habitat suitability for *Anodorhynchus hyacinthinus* based on the ensemble model, combining the BIOCLIM, MAXENT, GAM, and GLM methods

4. Discussion

The fluctuation between daily and nighttime temperatures, along with precipitation during the wettest month, were the most important predictive variables in determining the forecasts of the generated models. All models indicate a scenario of low habitat suitability, considering climatic variables, for the states of Rio Grande do Sul, Santa Catarina, the extreme north of the Amazon, and much of the Brazilian coastline. For several other regions of the country, the models suggest acceptable habitat suitability, which implies that, potentially, the species could occur in a larger area than currently known. The continuity of the potential distribution area suggests, albeit indirectly, that the populations may not be entirely isolated, or if they are, this is likely not due to the presence of an inhospitable matrix of environmental variables, but possibly due to other factors such as human impact and biotic interactions.

Such models have many limitations; for example, they do not take into account the presence of cities or anthropogenic impacts on populations, or even the occurrence of species that the birds typically feed on. These factors can significantly restrict the areas predicted as suitable. However, for unsuitable areas, there is little doubt. The projections clearly show that habitats in coastal regions and the far southern part of the country are poorly suited for the species. This information could be very useful for the development of management plans or the practical relocation of the species. Additionally, the potential distribution areas highlighted by the models provide important insights for future conservation efforts. The identification of regions that may have been previously overlooked can aid in the prioritization of conservation actions, such as habitat restoration and protection. Moreover, these findings can guide future field studies to further validate the model predictions and assess the actual presence of the species in these newly identified regions, helping to refine conservation strategies and improve the species' long-term survival prospects.

5. Conclusions

The essence of this research lies in the development and application of innovative multi-temporal species distribution models to map and interpret the changing habitat of the Hyacinth Macaw in response to extensive land use transformation. Unlike previous studies, our approach combined year-specific occurrence data with dynamic land cover information, revealing both range expansion and increasing fragmentation depending on the biome. This method stands out for its ability to detect temporal trends and spatial priorities for conservation, even when occurrence data are sparse. The results demonstrate that while the species has adapted to some habitat changes, ongoing agricultural and pasture expansion threaten the long-term viability of its populations, especially given the low proportion of suitable habitat under formal protection. The distinctive insights provided by this study underscore the urgent need for targeted conservation actions, including the expansion of protected areas and restoration of native vegetation, to safeguard the future of the Hyacinth Macaw. The observed distributional shifts are explained by the interplay between natural habitat availability and human-driven land cover change, emphasizing the critical role of landscape management in species conservation.

Conflict of interest

I declare that I have no conflict of interest in relation to this research, whether financial, personal, authorship or otherwise, that could affect the research and its results presented in this paper.

Financing

The study was performed without financial support.

Data availability

Data will be made available on reasonable request.

Use of artificial intelligence

The author confirms that he did not use artificial intelligence technologies when creating the current work.

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